

19 — Two Aux Engines: the only scenario that exercises Level 2

Proves · The only scenario where Level 2 does visible work — and Level 3 with it.

Mechanics this scenario turns on

- Level 2 redistributes the hotel load between the shaft generator and the auxiliaries, looking for a cheaper split. It can only find one when there is something to redistribute — with the SG at its ceiling and a single aux, it returns zero.
- Level 3 (DRC) damps the hotel/mission swing: $\text{variation} \times 0.8$, minus whatever the battery already shaved, so the same kilowatt is never counted twice.
- Level 2 and Level 3 are computed for **Transit only** (decision D4/Q5 — no workbook counterpart elsewhere). Other modes get an empty result, and the pinned-baseline radio does not reach them.
- The shaft generator is filled before any auxiliary starts (the main engine is already turning), and its output is a load **on** that main engine — which is why the ME figure exceeds propulsion. SG capacity scales with the number of running MEs.
- Baseline rule: **no battery** → **the worst combination** ($\text{count} - 1$); **battery active** → **the third from worst** ($\text{Math.Max}(0, \text{count} - 3)$). It models what the ship is assumed to do today.

Panels described below · Why Level 2 was invisible · The plant, built for exactly that · Level 1 · Level 2 — the point of the scenario · Result · Open question for the workbook

Anything not described here — the mechanics above name the step that produced it; **00-ORIENTATION** Part 6 has the full number-to-step index.

Trust · characterisation snapshot, generated from the code. It detects change; it does NOT prove correctness. Figures marked *pending reference verification* have never been checked against anything outside the application.

Read after · scenario 11.

Why it exists. Level 2 produced **zero savings in all 18 original scenarios**. A whole optimization level — the recursive sweep — had no end-to-end coverage: break it so it returns zero and every snapshot would still pass.

Why Level 2 was invisible

Read the aux SFOC curve (engine 8) and it falls monotonically:

10 % → 228.47	50 % → 198.07	74–80 % → 193.00	100 % → 194.63
---------------	---------------	------------------	----------------

More load is always better. Level 1 already evaluates every engine **count** and picks the cheapest, so "run fewer engines harder" is a decision Level 1 has made before Level 2 is asked.

Level 2's only remaining lever is an **asymmetric** split — and that needs a demand a single engine cannot legally carry.

The plant, built for exactly that

ME 2 × 8 500 · no SG · AE 2 × 2 000 · propulsion 8 000 · hotel 2 000 · 5 000 h

- No shaft generator ⇒ the aux side must carry the whole 2 000 kW hotel load.
- 1 aux engine ⇒ $2\,000 / 2\,000 = 100\%$ — above the 90 % ceiling, rejected.
- 2 aux engines ⇒ 50 % each. Legal, and squarely in the steep part of the curve.

Level 1

Two valid combinations (ME = 0 is invalid in Transit, aux count is forced to 2):

#	ME	SG	AE	FOC t/h
0	2	off	2	1.734727 ← optimal
1	1	off	2	1.772322 ← baseline

Baseline **8 861.6 t/yr** · L1 savings **187.97 t/yr**.

Level 2 — the point of the scenario

Level 1 handed the aux side an even split: $2 \times 1\,000\text{ kW} = 50\%$ each. The sweep searches 10 %...90 % in 2 % steps and finds:

AE₁ 200 kW @ 10 % sfoc 228.47
AE₂ 1 800 kW @ 90 % sfoc 193.78

Hand-check

even split : 2 000 kW × 198.07 g/kWh	= 0.396140 t/h
asymmetric : 200 × 228.47 + 1 800 × 193.78	= 0.394498 t/h
difference × 5 000 h	≈ 8.21 t/yr
reported	= 8.14 t/yr

Agreement to within rounding of the curve read-off. **The exact figure is pending reference verification** — it came from the code, not the workbook.

Result

L1	187.97
L2	8.14
L3	45.25
Advanced / Pro / Premium	187.97 / 196.11 / 241.36

The first scenario in the suite where all three tiers differ.

Open question for the workbook

The model's answer is one generator idling at 10 % and one at 90 %. A crew would more likely run a **single** generator at 100 % — which the 90 % ceiling forbids. If the workbook disagrees with 10/90, the floor or the ceiling is mis-set. Worth confirming before this figure is treated as correct.

Takeaway: Level 2 only has room when a single engine cannot legally carry the aux demand. That is a narrow window, and no scenario had ever landed in it.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-07 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)